

Practice Problems for Hypothesis Statements

September 2026

There will be two statistics that you will be dealing with. One of them is the population mean which we denote with μ (read as mu, with a long “u”). The other is the population proportion which we will denote with π (common is p but that is inconsistent with the “Population parameters are written in Greek” rule of thumb).

For each scenario please correctly identify the null hypothesis, denoted H_0 , and the alternative hypothesis, denoted H_A . The first two questions are multiple choice to give you the idea. Remember that all possible situations (eg the entire real number line) has to be in either the null or alternative hypothesis.

HINT 1: The “equal” part will always go with the null hypothesis in this class and does generally in the vast majority of modern research. The only time it doesn’t, to the best of my knowledge, is equivalency testing.

HINT 2: The $\hat{\cdot}$ (read “hat”) in statistics means “estimated”. So the estimated proportion is $\hat{\pi}$ for example.

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1. A large company recently hired an advertising agency. The CEO of the company is interested in if more people recognize the company name now that the advertising is being ran from an outside company. The old proportion of people who recognized the company was .47. What is the correct Null Hypothesis and Alternative Hypothesis?

H_0

- A. $\pi \leq .47$
- B. $\hat{\pi} = .47$
- C. $\pi = .47$
- D. $\hat{\pi} = p$

H_A

- A. $\hat{\pi} > .47$
- B. $\pi > .47$
- C. $\pi \neq .47$
- D. $\pi \geq .47$

2. I happen to drink a lot of tea. About 6 cups in a day actually and occasionally some coffee to top it off (which is significantly down from when I originally wrote this problem in grad school which had it at 9 cups of tea and “commonly”). I get a lot of my tea down in the East Village of Des Moines at Gung Fu Tea where I generally buy tea in 4oz packages. Because of the amount of tea I buy from them I was once concerned they were, on average, underfilling the packages of tea. So over the course of a year I recorded the weight of tea in each 4oz package I bought on my own scales at home. After the year was over I found the mean weight of tea packages that I had bought. Please state the Null Hypothesis and Alternative Hypothesis. Note we are using μ for this problem because we are talking about averages/means.

H_0

- A. $\mu \geq 4$
- B. $\hat{\mu} = 4$
- C. $\mu = 4$
- D. $\hat{\mu} = \bar{y}$

H_A

- A. $\hat{\mu} < 4$
- B. $\mu < 4$
- C. $\mu \neq 4$
- D. $\mu \geq 4$

3. A business owner is interested to see if the a new manufacturer of sweatshirts is producing the sweatshirts at a better quality than the old manufacturer. To this end, the owner has a load of sweatshirts inspected to find the proportion that are defective. The old manufacturer had a defective proportion of about .04. Please state the Null Hypothesis and Alternative Hypothesis.

4. There is a professor at the University of Maine (Go Black Bears!!!) who is infamous for having his classes run over time. In a given semester an average UMaine professor would run over time with a proportion of about .10. Students in his class, after being taught hypothesis testing for proportions, recorded the proportion of days he ran over time and then preformed a hypothesis test. Please state the Null Hypothesis and Alternative Hypothesis.

5. Let's tweak the above question: There is a professor at the University of Maine who is infamous for having his classes run over time. In a given semester an average UMaine professor would run over time by an average of about 1 minute. Students in his class, after being taught hypothesis testing for means, recorded the time he ran over through the course of the semester and then preformed a hypothesis test. Please state the Null Hypothesis and Alternative Hypothesis.

6. A teacher in a middle school is interested to see if a new method of teaching reading would improve the literacy rate of their students. So for one year they taught with the new method and then looked at the standardized test scores for their students. From past standardized testing, the teacher knew that around 82% of their students would pass. Please state the Null Hypothesis and Alternative Hypothesis.

7. Cyride (bus system in Ames IA) tries to be on time as much as possible and to this end management established an aggressive new policy. After the policy went into effect, management wanted to know if the busses were delayed because too many bus stops had people running to catch the bus. To this end the drivers were instructed to keep a tally for how many bus stops they had to wait for "runners". The old scheduling method assumed the busses would wait at a rate of 1 in 5 bus stops. Please state the Null Hypothesis and Alternative Hypothesis. NOTE: This is a made up scenario as CyRide bus drivers would never wait. At times they would make eye contact with you as they drove away.